Current approaches to modelling the **spread** of **wildland fire**: a **review**

GLW Perry - Progress in Physical Geography, 1998 - journals.sagepub.com

… **review** considers the development of some of the models and modelling approaches designed to predict the **spread** and spatial behaviour of **wildland fire** … for **wildland fire management** …

☆ Save 99 Cite Cited by 276 Related articles All 11 versions

Machine learning and deep learning for **wildfire spread prediction**: A **review**

HS Andrianarivony, MA Akhloufi - Fire, 2024 - mdpi.com

… highly performant in predicting fire **behavior**. This paper presents a systematic **review** of recent ML and DL techniques developed for **wildfire spread prediction**, detailing the commonly …

☆ Save 99 Cite Cited by 19 Related articles All 4 versions »

Trending and emerging prospects of physics-based and ML-based **wildfire spread** models: A comprehensive **review**

H Singh, LM Ang, T Lewis, D Paudyal, M Acuna… - Journal of Forestry …, 2024 - Springer

… aiding in the development of more accurate and reliable **prediction** tools. Methods for literature **review** This **wildfire spread model review** analysed research published in peer-reviewed …

☆ Save 99 Cite Cited by 26 Related articles All 7 versions

A comparative **review** on **wildfire** simulators

GD Papadopoulos, FN Pavlidou - IEEE systems Journal, 2011 - ieeexplore.ieee.org

… This project promises to develop more reliable **bushfire spread** simulation and animation … In this paper, a **review** on **wildfire simulation** models has been presented. Before that, a brief …

☆ Save 99 Cite Cited by 173 Related articles All 9 versions

Uncertainty and risk in **wildland fire management**: a **review**

MP Thompson, DE Calkin - Journal of environmental **management**, 2011 - Elsevier

… into the impacts of wildfires on human life and property, ecosystems, and other valued resources, and into the factors contributing to **wildfire** occurrence and **spread**. Managing …

☆ Save 99 Cite Cited by 385 Related articles All 15 versions

A systematic **review** and bibliometric analysis of **wildland fire behavior** modeling

J Silva, J Marques, I Gonçalves, R Brito, S Teixeira… - Fluids, 2022 - mdpi.com

… a mathematical percept to **model** the **spread** of **wildland fire**. Bakhshaii and Johnson [7], in a recent **review**, called these last two models a first-generation type of **wildfire** models. The …

☆ Save 99 Cite Cited by 24 Related articles All 6 versions »

A **review** of a new generation of **wildfire**–atmosphere modeling

A Bakhshaii, EA Johnson - Canadian Journal of Forest …, 2019 - cdnsciencepub.com

… The earliest physical science models tend to deal with isolated microscale phenomena (eg, **wildfire spread model** or the optical properties of ice crystals) so that their direct application …

☆ Save 99 Cite Cited by 113 Related articles All 12 versions

A **review** of **wildland fire spread** modelling, 1990-present 3: Mathematical analogues and **simulation** models

AL Sullivan - arXiv preprint arXiv:0706.4130, 2007 - arxiv.org

… involving **wildland fire spread** that are not based on **wildland fire** … to the **spread** of fires and thus are described as **wildland fire spread** models, while in some cases **wildland fire spread** is …

☆ Save 99 Cite Cited by 51 Related articles All 6 versions »

A **review** of machine learning applications in **wildfire** science and **management**

P Jain, SCP Coogan, SG Subramanian… - Environmental …, 2020 - cdnsciencepub.com

… For example, in the task of modeling **wildland fire spread**, the relevant physical properties include fuel composition, local weather, and topography. The current state-of-the-art method in …

☆ Save 99 Cite Cited by 806 Related articles All 21 versions

A Comprehensive **Review** of Empirical and Dynamic **Wildfire** Simulators and Machine Learning Techniques used for the **Prediction** of **Wildfire** in Australia

H Singh, LM Ang, D Paudyal, M Acuna… - Technology, Knowledge …, 2025 - Springer

… better anticipate and mitigate the impacts of **wildfire spread**. This **review** will discuss the range of empirical and dynamic tools used for **wildfire prediction**, highlighting their advantages, …

☆ Save 99 Cite Cited by 4 Related articles All 4 versions

A **review** of recent advances in risk analysis for **wildfire management**

C Miller, AA Ager - International journal of **wildland fire**, 2012 - CSIRO Publishing

… We also **review** recent advances in addressing temporal **dynamics** of fire risk and spatial optimisation of fuels **management** … for future research and development in **wildfire** risk analysis. …

☆ Save 99 Cite Cited by 348 Related articles All 11 versions

A **review** of **wildland fire spread** modelling, 1990-present, 1: Physical and quasi-physical models

AL Sullivan - arXiv preprint arXiv:0706.3074, 2007 - arxiv.org

… Whereas FIRETEC was intended to be used to **model wildland fire spread** across large spatial scales, IUSTI concentrates on resolving the chemical and conservation equations at a …

☆ Save 99 Cite Cited by 62 Related articles All 4 versions »

Mathematical models and calculation systems for the study of **wildland fire** behaviour

E Pastor, L Zárate, E Planas, J Arnaldos - Progress in Energy and …, 2003 - Elsevier

This is a **review** of the most important work in **wildland fire** mathematical modelling which … **wildland fire** models to be sorted. Surface fire **spread** models, crown fire initiation and **spread** …

☆ Save 99 Cite Cited by 546 Related articles All 10 versions

Operational **wildfire** suppression modelling: a **review** evaluating development, state of the art and future directions

TJ Duff, KG Tolhurst - International Journal of **Wildland Fire**, 2015 - CSIRO Publishing

… **review**, we focus on the development and use of models that emulate fire suppression for decision support. We **review** … There are two main approaches to dynamic fire **spread simulation**…

☆ Save 99 Cite Cited by 92 Related articles All 7 versions

Wildland surface fire **spread** modelling, 1990–2007. 3: **Simulation** and mathematical analogue models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… The present series of articles endeavours to comprehensively **survey** and précis all types of surface fire **spread** models developed during the period 1990–2007. The present paper …

☆ Save 99 Cite Cited by 464 Related articles All 5 versions

Modelling smoke transport from wildland fires: a **review**

SL Goodrick, GL Achtemeier, NK Larkin… - … of **Wildland Fire**, 2012 - CSIRO Publishing

… Another grid **model** receiving attention for **wildland fire** related air quality issues is the … of the modelling studies cited in this **review** involves the **dynamics** of the buoyant phase of the …

☆ Save 99 Cite Cited by 149 Related articles All 8 versions

Limitations on the accuracy of **model** predictions of **wildland fire** behaviour: a state-of-the-knowledge overview

ME Alexander, MG Cruz - The forestry chronicle, 2013 - pubs.cif-ifc.org

… **model** predictions of **wildland fire** behaviour and includes a short historical overview of **wildland fire** behaviour research as it relates to **model** … Estimating **wildland fire** rate of **spread** in a …

☆ Save 99 Cite Cited by 130 Related articles All 8 versions

DEVS-FIRE: Towards an integrated **simulation** environment for surface **wildfire spread** and containment

L Ntaimo, X Hu, Y Sun - **Simulation**, 2008 - journals.sagepub.com

… applied to an example of fire **spread simulation** [44]. A recent … the **wildfire spread** and containment **simulation model** presented in … We **review** the basics of **wildfire behavior** [56] to set …

☆ Save 99 Cite Cited by 122 Related articles All 9 versions

Wildfire management in Canada: **Review**, challenges and opportunities

C Tymstra, BJ Stocks, X Cai, MD Flannigan - Progress in Disaster Science, 2020 - Elsevier

… **management** agencies to allow more **wildfire** on the landscape by implementing a risk-based … future path for **wildfire management** in Canada. To co-exist with **wildfire**, agencies in …

☆ Save 99 Cite Cited by 339 Related articles All 6 versions

Wildland surface fire **spread** modelling, 1990–2007. 1: Physical and quasi-physical models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… readers, and this is the main reason for the inclusion of his work in the current **review**. … Whereas FIRETEC was intended to be used to **model wildland fire spread** across large spatial …

☆ Save 99 Cite Cited by 630 Related articles All 5 versions

Wildland surface fire **spread** modelling, 1990–2007. 2: Empirical and quasi-empirical models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… **spread** for the prevailing conditions (the **prediction** of which is the primary aim of all fire **spread prediction** … covered in the present **review** as it is outside the scope of surface fire **spread**). …

☆ Save 99 Cite Cited by 583 Related articles All 4 versions

Wildfire prediction to inform fire **management**: statistical science challenges

SW Taylor, DG Woolford, CB Dean, DL Martell - 2013 - projecteuclid.org

… In this paper we **review** some of the models that have been developed to predict fire occurrence, … **spread model** against **wildfire** observations. A major challenge in this regard is that …

☆ Save 99 Cite Cited by 195 Related articles All 13 versions

Evaluating a **model** for predicting active crown fire rate of **spread** using **wildfire** observations

ME Alexander, MG Cruz - Canadian Journal of Forest Research, 2006 - cdnsciencepub.com

… With 57 quantitative **wildfire** observations, no study to date … to evaluate a predictive fire **behavior model**. Thus, contrary to … — A **review** of fire weather, fuels, and fire **behavior**. Can. For. …

☆ Save 99 Cite Cited by 149 Related articles All 14 versions

Review of **wildfire management** techniques—Part I: Causes, prevention, detection, suppression, and data analytics

S Jazebi, F De Leon, A Nelson - IEEE Transactions on Power …, 2019 - ieeexplore.ieee.org

… in consecutive frames over time for remote monitoring of **wildfire spread**. In [45], a … **forest fire** with low consumption of energy. The fire **spread model** in [46] can produce fire **propagation** …

☆ Save 99 Cite Cited by 145 Related articles All 2 versions

On the need for a theory of **wildland fire spread**

MA Finney, JD Cohen, SS McAllister… - … journal of **wildland fire**, 2012 - CSIRO Publishing

… In summary, our **review** of existing research indicates an insufficient experimental basis for … an insufficient basis for related fire **spread model** assumptions. Independent experimental …

☆ Save 99 Cite Cited by 312 Related articles All 12 versions

Modeling **wildland fire propagation** with level set methods

V Mallet, DE Keyes, FE Fendell - Computers & Mathematics with …, 2009 - Elsevier

… burning, and post-crisis **review**, eg, personnel training, … **wildland fire spread** models, especially a semi-empirical, equilibrium-type **model** proposed in [11] for wind-aided fire **spread** …

☆ Save 99 Cite Cited by 143 Related articles All 17 versions

A **review** on status, implications and recent trends of **forest fire management**

V Attri, R Dhiman, S Sarvade - Archives of Agriculture and …, 2020 - academia.edu

… **Forest fire spread** out in an area having combustible material … **Forest fire** starts from the ignition point and **spread** outwards … topography were made favorable environment for **forest fire**. …

☆ Save 99 Cite Cited by 65 Related articles All 6 versions »

A **review** of recent forest and **wildland fire management** decision support systems research

DL Martell - Current Forestry Reports, 2015 - Springer

… Fire behaviour or the physics and chemistry of fire ignition and **spread**. … focus of this **review** is very decidedly North America for the period 2010 through 2014. **Wildland fire** and the …

☆ Save 99 Cite Cited by 167 Related articles All 5 versions

Forest fire spread simulation and fire extinguishing visualization research
Q Meng, H Lu, Y Huai, H Xu, S Yang - Forests, 2023 - mdpi.com
… **forest fire spread** quickly. Therefore, this paper proposes a lightweight **forest fire spread model** … ensuring the realism of **forest fire behavior** and the immersion of the virtual environment. …
☆ Save 99 Cite Cited by 21 Related articles All 5 versions »

Forest fire spread simulating **model** using cellular automaton with extreme learning machine
Z Zheng, W Huang, S Li, Y Zeng - Ecological Modelling, 2017 - Elsevier
… **simulation** of **forest fire** spreading plays an essential role in designing quick risk **management** and … By integrating the ELM with the traditional **forest fire** CA framework, a new cellular …
☆ Save 99 Cite Cited by 135 Related articles All 8 versions

WRF-Fire: coupled weather–**wildland fire** modeling with the weather research and forecasting **model**
JL Coen, M Cameron, J Michalakes… - Journal of Applied …, 2013 - journals.ametsoc.org
… numerical weather **prediction model**. The fire module is a surface fire-**behavior model** that is two-way coupled with the atmospheric **model**. Near-surface winds from the atmospheric …
☆ Save 99 Cite Cited by 347 Related articles All 15 versions

Wildland fire spread modeling using convolutional neural networks
JL Hodges, BY Lattimer - Fire technology, 2019 - Springer
… **spread** requires multiple simulations that may limit the number of scenarios which can be analyzed during an emergency. A new **wildland fire spread model** … **wildland fire spread model** …
☆ Save 99 Cite Cited by 149 Related articles All 5 versions

Traffic modeling for wildland–urban interface fire evacuation
P Intini, E Ronchi, S Gwynne, A Pel - Journal of Transportation …, 2019 - ascelibrary.org
… Based on this **review** analysis, recommendations are made … possible integrations with **wildfire** and pedestrian models. … and fire **propagation** speed may not affect the **model** choice. …
☆ Save 99 Cite Cited by 90 Related articles All 11 versions

A method for ensemble **wildland fire simulation**
MA Finney, IC Grenfell, CW McHugh, RC Seli… - … Modeling & Assessment, 2011 - Springer
… An ensemble **simulation** system that accounts for uncertainty in long-range weather conditions and two-dimensional **wildland fire spread** is described. Fuel moisture is expressed based …
☆ Save 99 Cite Cited by 267 Related articles All 11 versions

Review of **wildfire** modeling considering effects on land surfaces
D Or, E Furtak-Cole, M Berli, R Shillito… - Earth-Science …, 2023 - Elsevier
… **review** are: (1) to introduce hydrologists, soil scientists and ecologists to principles and advances in modeling of **wildfire dynamics** and rates of **spread** … large eddy **simulation** models. We …
☆ Save 99 Cite Cited by 43 Related articles All 3 versions

Review of vortices in **wildland fire**
JM Forthofer, SL Goodrick - Journal of Combustion, 2011 - Wiley Online Library
… fire **behavior** and safety concerns. In this paper, the current state of knowledge of the interaction of **wildland fire** … In this study a computer **model** of the atmospheric boundary layer was …
☆ Save 99 Cite Cited by 136 Related articles All 28 versions

A cellular automaton **model** of **wildfire propagation** and extinction
KC Clarke, JA Brass, PJ Riggan - … Engineering and Remote …, 1994 - research.fs.usda.gov
… We propose a new **model** to predict the spatial and temporal **behavior** of wildfires. Fire **spread** and intensity were simulated using a cellular automaton **model**. Monte Carlo techniques …
☆ Save 99 Cite Cited by 300 Related articles All 11 versions »

Emulation of **wildland fire spread simulation** using deep learning

F Allaire, V Mallet, JB Filippi - Neural networks, 2021 - Elsevier

… Numerical **simulation** of **wildland fire spread** is useful to … of the burned surface returned after **simulation** of a fire igniting … After training on a large **simulation** dataset, the network shows …

☆ Save 99 Cite Cited by 65 Related articles All 11 versions

Developing the US **wildland fire** decision support system

EK Noonan-Wright, TS Opperman… - Journal of …, 2011 - Wiley Online Library

… , fire **behavior prediction**, economic assessment, smoke … The objective of this paper is to **review** the US **Wildland Fire** … environment in the context of **wildland fire management**. Next, we …

☆ Save 99 Cite Cited by 180 Related articles All 8 versions

AI for **Wildfire Management**: From **Prediction** to Detection, **Simulation**, and Impact Analysis—Bridging Lab Metrics and Real-World Validation

N Caron, HN Noura, L Nakache, C Guyeux, B Aynes - AI, 2025 - mdpi.com

… In this article, we **review** the main domains of **wildfire management** where AI has been … We first identify high-risk regions, **forecast** risk, detect predicted ignitions, simulate fire **spread**, and …

☆ Save 99 Cite Related articles All 2 versions »

An overview of mountain meteorological effects relevant to fire behaviour and **bushfire** risk

JJ Sharples - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… The likelihood of growth of a **bushfire** is typically derived using some form of fire **spread model** that … risk in the Australian high country, it is prudent to begin with a **review** of the available …

☆ Save 99 Cite Cited by 195 Related articles All 12 versions

A **Survey** on Anomaly detection in Evolving Data: [with Application to **Forest Fire** Risk **Prediction**]

M Salehi, L Rashidi - ACM SIGKDD Explorations Newsletter, 2018 - dl.acm.org

… **model** hence the spatiotemporal customisation of the **model** cannot be assessed. In [31], it is claimed that **forest fire** risk needs to consider both **forest fire** … controlling **forest fire spread**, …

☆ Save 99 Cite Cited by 130 Related articles All 7 versions

Software-based simulations of **wildfire spread** and wind-fire interaction

M Ghodrat, F Shakeriaski, SA Fanaee, A Simeoni - Fire, 2022 - mdpi.com

… **review** will be on providing an overview of available **wildfire** simulators. The fire **behavior** models including available **simulation** software and studies that address **wildfire propagation** …

☆ Save 99 Cite Cited by 13 Related articles All 5 versions »

Social science research related to **wildfire management**: an overview of recent findings and future research needs

S McCaffrey, E Toman, M Stidham… - … Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… that many of the basic social **wildfire dynamics** are similar across countries. Of … **survey** results and **wildland fire management** implications. In ‘The public and **wildland fire management**…

☆ Save 99 Cite Cited by 333 Related articles All 13 versions

Estimating **wildfire behavior** and effects

FA Albini - 1976 - books.google.com

… those concerned with **wildland fire behavior** and effects. Although fire **behavior prediction** is by … Here we **review** some **model** uses and indicate the type of output needed and the general …

☆ Save 99 Cite Cited by 1324 Related articles All 10 versions

SPARK–A **bushfire spread prediction** tool

C Miller, J Hilton, A Sullivan, M Prakash - International Symposium on …, 2015 - Springer

… system for bushfire **spread prediction**, SPARK, which was developed with the functionality to **model** some of these … Submitted to Environmental Modelling and Software (under **review**) …

☆ Save 99 Cite Cited by 127 Related articles All 7 versions

Defining extreme **wildfire** events: Difficulties, challenges, and impacts

F Tedim, V Leone, M Amraoui, C Bouillon… - Fire, 2018 - mdpi.com

… on measurable fire **spread** and **behavior** parameters and … This paper will first present a **review** of the terms used to label … organizations concerned with **wildfire management** but also with …

☆ Save 99 Cite Cited by 526 Related articles All 21 versions »

Spatial **prediction** of **wildfire** susceptibility using field **survey** GPS data and machine learning approaches

O Ghorbanzadeh, K Valizadeh Kamran, T Blaschke… - Fire, 2019 - mdpi.com

… A **wildfire** data inventory was created by integrating global …) approaches for the spatial **prediction** of **wildfire** susceptibility. The … All ML approaches were trained using 75% of the **wildfire** …

☆ Save 99 Cite Cited by 179 Related articles All 13 versions »

DEVS-FIRE: design and application of formal discrete event **wildfire spread** and suppression models

X Hu, Y Sun, L Ntaimo - **Simulation**, 2012 - journals.sagepub.com

… Next we discuss how the fire **spread model** and fire suppression **model** work, and provide formal … Current approaches to modeling the **spread** of **wildland fire**: A **review**. Prog Phys …

☆ Save 99 Cite Cited by 84 Related articles All 5 versions

Human-caused fire occurrence modelling in perspective: a **review**

S Costafreda-Aumedes, C Comas… - … of **Wildland Fire**, 2017 - CSIRO Publishing

… process modelled separately from fire **spread**, which may or … global **wildfire** databases, novel **model** development, some … for improved transference to **management**. Fire occurrence …

☆ Save 99 Cite Cited by 206 Related articles All 14 versions

Guidance on spatial **wildland fire** analysis: models, tools, and techniques

RD Stratton - 2006 - books.google.com

… crown fire **behavior** advances, so too will our fire **management** applications" (Scott … **wildland fire** analysis. It is a product of seven years of experience in support of incident **management**, …

☆ Save 99 Cite Cited by 132 Related articles All 6 versions

A **review** of approaches to estimate **wildfire** plume injection height within large-scale atmospheric chemical transport models

R Paugam, M Wooster, S Freitas… - … Chemistry and Physics, 2016 - acp.copernicus.org

… we cannot resolve the plume **dynamics**, parameterizations are … The aim of this manuscript is to **review** the different … of the Sofiev **model** against (i) the original version of the **model**, (ii) …

☆ Save 99 Cite Cited by 187 Related articles All 22 versions »

Wildfire likelihood's elements: a literature **review**

M Mhawej, G Faour, J Adjizian-Gerard - Challenges, 2015 - mdpi.com

… It shall contribute to the development of sustainable forest **management** plans. Similarly, this … that affect the ignition of the **wildfire**, its initial **spread** and its free **propagation**). The first set …

☆ Save 99 Cite Cited by 48 Related articles All 10 versions »

Overview of **forest fire propagation** research

DX Viegas - Fire Safety Science, 2011 - publications.iafss.org

… socio-economical aspects, **management** and consequences of forest fires … **propagation**. In the opinion of the present author, fire **spread** or **propagation** is a central issue of the **forest fire** …

☆ Save 99 Cite Cited by 21 Related articles All 3 versions »

Integrating fire **behavior** models and geospatial analysis for **wildland fire** risk assessment and fuel **management** planning

AA Ager, NM Vaillant, MA Finney - Journal of Combustion, 2011 - Wiley Online Library

… This work was initiated in 2004 with an ad hoc **review** of existing fire **behavior** modeling … **wildland fire spread** models described in this paper will contribute to improved **wildfire simulation** …

☆ Save 99 Cite Cited by 170 Related articles All 8 versions

Global pyrogeography: the current and future distribution of **wildfire**

MA Krawchuk, MA Moritz, MA Parisien, J Van Dorn… - PloS one, 2009 - journals.plos.org

… **wildfire**, suggesting severely altered fire regimes and the need for more explicit inclusion of fire in research on global vegetation-climate change **dynamics** … .1 historical **simulation** data, …

☆ Save 99 Cite Cited by 1144 Related articles All 25 versions »

Community **wildfire** preparedness: A global state-of-the-knowledge summary of social science research

S McCaffrey - Current Forestry Reports, 2015 - Springer

… The original synthesis had a broad focus that included internal fire **management dynamics**.
… included in this **review** which focuses solely on public response to **wildfire management**. A …

☆ Save 99 Cite Cited by 227 Related articles All 6 versions

Empirical-based models for predicting head-fire rate of **spread** in Australian fuel types

MG Cruz, JS Gould, ME Alexander, AL Sullivan… - Australian …, 2015 - Taylor & Francis

… of each **model** of head-fire rate of **spread** and their … -fire rate of **spread** described in this **review** are listed in Table … In presenting the various **bushfire** rate of **spread** models used in …

☆ Save 99 Cite Cited by 120 Related articles All 5 versions

Studying **wildfire behavior** using FIRETEC

R Linn, J Reisner, JJ Colman… - … journal of **wildland fire**, 2002 - CSIRO Publishing

… capacity for a physics-based **model** capable of describing **wildfire behavior** in a wide variety of circumstances. We believe that a three-dimensional **wildfire model** of this type could help …

☆ Save 99 Cite Cited by 569 Related articles All 5 versions

Climate change impact on future **wildfire** danger and activity in southern Europe: a **review**

J Dupuy, H Fargeon, N Martin-StPaul, F Pimont… - Annals of Forest …, 2020 - Springer

… indices describe the potential for fire **spread** (the Initial **Spread** Index, ISI), … **spread** in SPITFIRE, while the maximum daily Nesterov index is used to calibrate the annual burnt area **model** …

☆ Save 99 Cite Cited by 401 Related articles All 20 versions

A two-dimensional reaction-advection-diffusion **model** of the **spread** of fire in wildlands

P Grasso, MS Innocente - 2018 - ap1.sib.uc.pt

… The aim of this research is to develop a **model** of the **spread** of wildfires … The **model** consists of two coupled partial differential … The aim of this research is to **model** the **spread** of **forest fire** …

☆ Save 99 Cite Cited by 16 Related articles All 3 versions »

Atmospheric interactions with **wildland fire** behaviour–I. Basic surface interactions, vertical profiles and synoptic structures

BE Potter - International Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… rate and wind speed based on the results of their numerical **model** with coupled fire … Part II of this **review** discusses plume structure and **dynamics** in detail, including elaboration on …

☆ Save 99 Cite Cited by 136 Related articles All 5 versions

Estimation of **wildfire** size and risk changes due to fuels treatments

MA Cochrane, CJ Moran, MC Wimberly… - … of **Wildland Fire**, 2012 - CSIRO Publishing

… modelling system (Finney 2004) to simulate the observed **wildfire** progression and **spread** rates. **Model** simulations were parameterised with LANDFIRE fuels and topographic data …

☆ Save 99 Cite Cited by 196 Related articles All 13 versions

Development and structure of Prometheus: the Canadian **wildland fire** growth **simulation model**

C Tymstra, RW Bryce, BM Wotton, SW Taylor… - … Report NOR-X-417 …, 2010 - spyd.com

… • to **review** and evaluate existing models; • to … a **wildfire** at many points around the fire perimeter, including the underlying rates of **spread** for the spatial **simulation** of firefront **propagation**. …

☆ Save 99 Cite Cited by 373 Related articles All 5 versions »

Distributed **wildfire** surveillance with autonomous aircraft using deep reinforcement learning

KD Julian, MJ Kochenderfer - … of Guidance, Control, and **Dynamics**, 2019 - arc.aiaa.org

… work together to monitor the **wildfire** growth. Furthermore, the … for both **wildfire propagation** and **trajectory** optimization is … **simulation**, the **wildfire** is small, and so no precise **trajectory** …

☆ Save 99 Cite Cited by 154 Related articles All 3 versions

A **review** of radiant heat flux models used in **bushfire** applications

AL Sullivan, PF Ellis, IK Knight - … Journal of **Wildland Fire**, 2003 - CSIRO Publishing

… The **prediction** of the thermal radiation from a **wildfire** is … is important in a number of fields, including rate of **spread** modelling, … of heat transfer and fire **propagation**, the physics is reason…

☆ Save 99 Cite Cited by 140 Related articles All 3 versions

Federal **forest-fire** policy in the United States

SL Stephens, LW Ruth - Ecological applications, 2005 - Wiley Online Library

… **wildfire** has increased over the last decade. Given the current and future challenges of fire **management**, and based on analytical research and **review** … not contribute to **wildfire spread** or …

☆ Save 99 Cite Cited by 836 Related articles All 18 versions

What do we know about **forest fire** size distribution, and why is this knowledge useful for forest **management**?

W Cui, AH Perera - International Journal of **Wildland Fire**, 2008 - CSIRO Publishing

… in forest **management** and **forest fire management**. In the present article, our goal is to **review** and … For example, slope affects rate of **spread** while aspect affects **spread** direction. …

☆ Save 99 Cite Cited by 120 Related articles All 6 versions

Analysis of evacuation **behavior** in a **wildfire** event

T Toledo, I Marom, E Grimberg, S Bekhor - International journal of disaster …, 2018 - Elsevier

… place, is critical for the efficient **management** of the event. The usual … A large-scale **wildfire** took place in Haifa on November 24, … Shortly after the fire events, a web **survey** was developed …

☆ Save 99 Cite Cited by 141 Related articles All 5 versions

LANDFIRE: a nationally consistent vegetation, **wildland fire**, and fuel assessment

MG Rollins - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… of different **wildfire management** strategies, **wildfire** season … ; (3) communications and iterative **review** by local ecologists and … the probability of individual **wildfire spread** and severity and …

☆ Save 99 Cite Cited by 791 Related articles All 14 versions

A simple physical **model** for **forest fire spread**

E Koo, P Pagni, J Woycheese… - In: Gottuk, D.; …, 2005 - research.fs.usda.gov

… mechanisms of **forest fire spread** and introduced a simple physics-based fire **spread model** for … Pitts [12] provided a thorough **review** of wind effects on fire. Weise and Biging [13] built a …

☆ Save 99 Cite Cited by 98 Related articles All 10 versions »

Modeling vorticity-driven **wildfire behavior** using near-field techniques

JJ Sharples, JE Hilton - Frontiers in Mechanical Engineering, 2020 - frontiersin.org

… An example **simulation** using the pyrogenic vector potential coupled to a **wildfire spread simulation** is shown in Figure 2 with s c = 0.5 and s w = 0.5 in Equation (8). These constants …

☆ Save 99 Cite Cited by 22 Related articles All 2 versions »

Evacuation decision-making and **behavior** in wildfires: Past research, current challenges and a future research agenda

E Kuligowski - Fire Safety Journal, 2021 - Elsevier

… an understanding of human **behavior** in wildfires, which is an … decision-making and **behavior** during wildland-urban … support developments of evacuation **simulation** models, and …

☆ Save 99 Cite Cited by 135 Related articles All 6 versions

Describing wildland surface fuel loading for fire **management**: a **review** of approaches, methods and systems

RE Keane - International Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… models are being developed, it is critical that new fuel classification systems be built to balance ecological understanding of fuel **dynamics** with the new input **model** requirements to …

☆ Save 99 Cite Cited by 194 Related articles All 10 versions

Current **wildland fire** patterns and challenges in Europe: A synthesis of national perspectives

N Fernandez-Anez, A Krasovskiy… - Air, Soil and Water …, 2021 - journals.sagepub.com

… involved in **wildland fire** research and land **management**, … to assess and understand fire regime, **dynamics**, and post-fire … fire **behavior**, linkages between weather and **wildland fire** …

☆ Save 99 Cite Cited by 166 Related articles All 52 versions

How risk **management** can prevent future **wildfire** disasters in the wildland-urban interface

DE Calkin, JD Cohen, MA Finney… - Proceedings of the …, 2014 - pnas.org

… First, we **review** how the success, or lack thereof, of **wildfire** … fire **behavior** in the nearby wildlands, meaning that proper care of the HIZ separates home losses from **wildland fire behavior**, …

☆ Save 99 Cite Cited by 639 Related articles All 23 versions

Negative consequences of positive feedbacks in US **wildfire management**

DE Calkin, MP Thompson, MA Finney - Forest Ecosystems, 2015 - Springer

… this article we **review** the structure of US **wildfire management** … ecological drivers of **wildfire management** response in both … the **trajectory** of **wildfire** risk and associated **management** …

☆ Save 99 Cite Cited by 397 Related articles All 28 versions

Forest fire dynamics in Nepal: Regional trends and socio-ecological drivers

KP Joshi, S Giri, D Kuinkel, S Kuikel, R Devkota… - Trees, Forests and …, 2025 - Elsevier

… national **forest fire** analyses. This study investigates fire trends, key drivers, and public perceptions of **management** … **Survey** responses highlight strong community support for awareness …

☆ Save 99 Cite Cited by 1 Related articles All 3 versions

Detecting critical nodes in forest landscape networks to reduce **wildfire spread**

D Yemshanov, N Liu, DK Thompson, MA Parisien… - PloS one, 2021 - journals.plos.org

… to plan **wildfire** fuel treatments that minimize the risk of fire **spread** in … **simulation** modeling to estimate the probability of fire **spread** … A **review** of operational research studies in **forest fire** …

☆ Save 99 Cite Cited by 22 Related articles All 10 versions »

Physical phenomena and length scales governing the behaviour of wildfires: a case for physical modelling

D Morvan - Fire technology, 2011 - Springer

… **review** concerning fire **spread** and fire behaviour models [9–11]. … as a “fully” physical **wildfire model** a **model** based on the solution of … For this **simulation** we found a rate of **spread** (ROS), …

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Evidence for lack of a fuel effect on forest and shrubland fire rates of **spread** under elevated fire danger conditions: implications for modelling and **management**

MG Cruz, ME Alexander… - … Journal of **Wildland Fire**, 2022 - CSIRO Publishing

… Our **review** and analysis of the relevant scientific … **spread model** outputs. We then look at the results from studies analysing **wildfire** occurrence data and observations of **wildfire spread** …

☆ Save 99 Cite Cited by 35 Related articles All 8 versions

A **survey** on technologies for automatic **forest fire** monitoring, detection, and fighting using unmanned aerial vehicles and remote sensing techniques

C Yuan, Y Zhang, Z Liu - Canadian journal of forest research, 2015 - cdnsciencepub.com

… and tracking of the **propagation** of large forest fires. A numerical **propagation model** for **forest fire** monitoring and detection was verified in **simulation** using a dynamic UAV **model** with six …

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Landscape–**wildfire** interactions in southern Europe: implications for landscape **management**

F Moreira, O Viedma, M Arianoutsou, T Curt… - … **management**, 2011 - Elsevier

… In this paper we **review** the available scientific knowledge on the … **wildfire**-landscape relationships in the Mediterranean region. Solid lines indicate the main drivers of fire spatial **spread** …

☆ Save 99 Cite Cited by 1066 Related articles All 31 versions

Forest fire propagation

D Xavier Viegas - … Transactions of the Royal Society of …, 1998 - royalsocietypublishing.org

… In the **management** of **forest fire** suppression tactics one is usually interested in predicting the overall time-evolution of the entire fire front. This is the so-called global range modelling …

☆ Save 99 Cite Cited by 133 Related articles All 4 versions

Getting ahead of the **wildfire** problem: Quantifying and mapping **management** challenges and opportunities

CD O'Connor, MP Thompson, F Rodríguez y Silva - Geosciences, 2016 - mdpi.com

… An important caveat to any potential changes to **wildfire management** … In this **review**, we provide an overview of some of the … Physical fire **behavior** and **spread** in the FSim **model** is also …

☆ Save 99 Cite Cited by 94 Related articles All 11 versions »

Socio-ecological perceptions of **wildfire management** and effects in Greece

P Palaiologou, K Kalabokidis, A Troumbis, MA Day… - Fire, 2021 - mdpi.com

… emphasis on reforming **wildfire management** policies to deal … of climate change in **wildfire** frequency and **behavior**. In addition… -based **survey** of professionals in **wildfire management** and …

☆ Save 99 Cite Cited by 44 Related articles All 11 versions »

FireCast: Leveraging Deep Learning to Predict **Wildfire Spread**.

D Radke, A Hessler, D Ellsworth - IJCAI, 2019 - ijcai.org

… areas surrounding a burning **wildfire** have high-risk of near-future **wildfire spread**, based on historical … to a random **prediction model** and a commonly used **wildfire spread model**, Farsite, …

☆ Save 99 Cite Cited by 165 Related articles All 4 versions »

A novel **wildfire** digital-twin framework using interactive **wildfire spread** simulator

K Hyeong-su, K Jin-Woo, S Yun… - … on Ubiquitous and …, 2019 - ieeexplore.ieee.org

… by supporting a **wildfire spread prediction**. The existing … **wildfire spread prediction**, but the **simulation** may cause the problems of high complexity and processing time [9]. The **simulation** …

☆ Save 99 Cite Cited by 30 Related articles

Efficient **forest fire** occurrence **prediction** for developing countries using two weather parameters

GE Sakr, IH Elhajj, G Mitri - Engineering Applications of Artificial …, 2011 - Elsevier

… A literature **review** of **forest fire prediction** methods divided … **prediction** systems of **forest fire** occurrence involving a large number of monitored features and comprising weather **prediction** …

☆ Save 99 Cite Cited by 130 Related articles All 3 versions

Simulation of **forest fire spread** based on a 3D-surface Cellular Automata **Model**

H Hua-guo, Z Xiao-li, W Lei - Journal of Beijing Forestry University, 2024 - j.bjfu.edu.cn

… the **forecast** and **simulation** visualization of **forest fire spread** the … (CA) to **forest fire** modelling and **spread simulation** based on … **Review** and prospect of researches on **forest fire** spreading …

☆ Save 99 Cite Cited by 19 Related articles »

A **review** of the main driving factors of **forest fire** ignition over Europe

A Ganteaume, A Camia, M Jappiot… - … **management**, 2013 - Springer

… , with increasing mobility of people, **spread** of tourism and recreational activities and the … , fire is a traditional and long established **management** tool. Its ancient and current use in …

☆ Save 99 Cite Cited by 732 Related articles All 23 versions

A **review** of the relationships between drought and **forest fire** in the United States

JS Littell, DL Peterson, KL Riley, Y Liu… - Global change …, 2016 - Wiley Online Library

… If drought continues, the number of days with elevated fuel flammability and fire **spread** … indices and fire **behavior** metrics have been used in the literature to **model** fire occurrence, …

☆ Save 99 Cite Cited by 616 Related articles All 15 versions

Wildfire and climate change adaptation of western North American forests: a case for intentional **management**

PF Hessburg, SJ Prichard, RK Hagmann… - Ecological …, 2021 - Wiley Online Library

… **review** the strength of evidence documenting changes or lack thereof. Similarly, in Prichard et al. (2021), we **review** … and temporal controls that drive **wildfire behavior** and effects, and the …

☆ Save 99 Cite Cited by 245 Related articles All 22 versions

Review of pathways for building fire **spread** in the wildland urban interface part I: exposure conditions

SE Caton, RSP Hakes, DJ Gorham, A Zhou, MJ Gollner - Fire technology, 2017 - Springer

… regions, proper forest **management** practices, such … **spread** (ROS) of a **wildland fire** is a relevant parameter for WUI purposes, both because it signals the rate at which a fire will **spread** …

☆ Save 99 Cite Cited by 284 Related articles All 7 versions

Data-driven **wildfire spread** modeling of european wildfires using a spatiotemporal graph neural network

M Rösch, M Nolde, T Ullmann, T Riedlinger - Fire, 2024 - mdpi.com

… yet applied STGNNs to **model** the daily **wildfire spread** on a … DL-based **wildfire spread model** to predict the daily **spread** of … uses a STGNN to **model** the **wildfire spread** for the first time. …

☆ Save 99 Cite Cited by 13 Related articles All 6 versions »

Wildfire spreading **prediction** using multimodal data and deep neural network approach

D Shadrin, S Illarionova, F Gubanov, K Evteeva… - Scientific reports, 2024 - nature.com

… critical issue in fire **behavior prediction**, so we analyze their contribution to the **model**’s results. … In this study, our focus is on predicting **wildfire spread** using multimodal data to assist in …

☆ Save 99 Cite Cited by 59 Related articles All 8 versions

A multi-fidelity framework for **wildland fire behavior** simulations over complex terrain

M Vanella, K McGrattan, R McDermott, G Forney… - Atmosphere, 2021 - mdpi.com

… A method for the large-eddy **simulation** (LES) of **wildfire spread** … A recent **review**, with an emphasis on operational … fire **behavior** between a **simulation** that models fire **spread** and heat …

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A **simulation** of probabilistic **wildfire** risk components for the continental United States

MA Finney, CW McHugh, IC Grenfell, KL Riley… - … Research and Risk …, 2011 - Springer

… To make the fire growth **simulation** efficient, fire **behavior** for the entire landscape was pre-… as precise as the centroid for the Public Land **Survey** System (PLSS) section in which the fire …

☆ Save 99 Cite Cited by 493 Related articles All 23 versions

Wildland fire emissions, carbon, and climate: Science overview and knowledge needs

WT Sommers, RA Loehman, CC Hardy - Forest Ecology and **Management**, 2014 - Elsevier

… **management** of forests for increased carbon sequestration, and because wildland fires are a key component of the carbon cycle, fire ecology, **behavior**, … cycle and **wildfire** and carbon …

☆ Save 99 Cite Cited by 119 Related articles All 13 versions

A **review** of the effects of **wildfire** smoke on the health and **behavior** of wildlife

OV Sanderfoot, SB Bassing, JL Brusa… - Environmental …, 2022 - iopscience.iop.org

… A conceptual **model** illustrating the various pathways by which **wildfire** smoke could influence wildlife populations and ecosystems. We hypothesize how effects of smoke on individual …

☆ Save 99 Cite Cited by 106 Related articles All 4 versions

Streamlined wildland-urban interface fire tracing (SWUIFT): modeling **wildfire spread** in communities

N Masoudvaziri, FS Bardales, OK Keskin… - … Modelling & Software, 2021 - Elsevier

… This paper studies the process of **wildfire spread** inside WUI communities. The fire **spread** … The new **model** is validated using the documented **spread** rates during the 2007 Witch and …

☆ Save 99 Cite Cited by 38 Related articles All 6 versions

cffdrs: an R package for the Canadian **forest fire** danger rating system

X Wang, BM Wotton, AS Cantin, MA Parisien… - Ecological …, 2017 - Springer

… on **wildland fire** ignition and **behavior** has been incorporated into the operational decision making within **wildland fire management** … **Forest Fire** Danger Rating System, into their **wildfire** …

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Assessment of deep learning models integrated with weather and environmental variables for **wildfire spread prediction** and a case study of the 2023 Maui fires

J Kim, Y Hu, N Elhami-Khorasani, K Sun… - arXiv preprint arXiv …, 2025 - arxiv.org

… -based fire **spread** models can … **spread prediction** based on over a decade of **wildfire** data in the state of Hawaii. We also compare deep learning models with a widely used non-AI **model**…

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Application of the socio-ecological system framework to **forest fire** risk **management**: A systematic literature **review**

I Vigna, A Besana, E Comino, A Pezzoli - Sustainability, 2021 - mdpi.com

… Envision in central and southcentral Oregon to evaluate the outcomes of current forest **management** decisions on future **forest fire dynamics**. In this case, a complex analysis of the …

☆ Save 99 Cite Cited by 53 Related articles All 12 versions »

Wildfirespreadts: A dataset of multi-modal time series for **wildfire spread prediction**

S Gerard, Y Zhao, J Sullivan - Advances in Neural …, 2023 - proceedings.neurips.cc

… of a **wildfire**. The only dataset we are aware of that represents **wildfire spread** predictions at a … In section 3, we describe related work in the field of **wildfire spread prediction**. We apply …

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Should we leave now? Behavioral factors in evacuation under **wildfire** threat

J McLennan, B Ryan, C Bearman, K Toh - Fire technology, 2019 - Springer

… research into **wildfire** evacuation **behavior** published between … In this paper we **review** published accounts of research … **simulation** or modeling research, (b) discussed only community …

☆ Save 99 Cite Cited by 176 Related articles All 11 versions

Updates and revisions to the 1992 Canadian **forest fire behavior prediction** system

BM Wotton, ME Alexander, SW Taylor - 2009 - researchgate.net

… This report documents a number of changes to the 1992 release of the Canadian **Forest Fire Behavior Prediction** (FBP) System, and addresses several mathematical and physical …

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The impact of fuel treatments on **wildfire behavior** in North American boreal fuels: a **simulation** study using FIRETEC

G Marshall, DK Thompson, K Anderson, B Simpson… - Fire, 2020 - mdpi.com

… to large wildfires that **spread** through natural forests… **wildfire** protection planning in forests not covered by existing operational fire **spread** models, we use FIRETEC to simulate fire **spread** …

☆ Save 99 Cite Cited by 36 Related articles All 11 versions »

A national approach for integrating **wildfire simulation** modeling into Wildland Urban Interface risk assessments within the United States

JR Haas, DE Calkin, MP Thompson - Landscape and Urban Planning, 2013 - Elsevier

… **wildfire** risk analysis is establishing the likelihood of **wildfire** … spatially resolved estimates of **wildfire** likelihood, which when … spatially-explicit **wildfire spread simulation model**. We classify …

☆ Save 99 Cite Cited by 146 Related articles All 12 versions

A **review** of prescribed burning effectiveness in fire hazard reduction

PM Fernandes, HS Botelho - International Journal of **wildland fire**, 2003 - CSIRO Publishing

… Obvious effects in **wildfire propagation** were observed in areas treated no more than 2–4 years … in the Canadian mixedwood boreal forest using two fire **behavior prediction** systems. …

☆ Save 99 Cite Cited by 935 Related articles All 14 versions

Progress towards and barriers to implementation of a risk framework for US federal **wildland fire** policy and decision making

DC Calkin, MA Finney, AA Ager, MP Thompson… - Forest Policy and …, 2011 - Elsevier

… we **review** progress towards the implementation of a risk **management** framework for US federal **wildland fire** … We first describe new developments in **wildfire simulation** technology that …

☆ Save 99 Cite Cited by 135 Related articles All 11 versions

Coupling wind **dynamics** into a DDDAS **forest fire propagation prediction** system

C Brun, T Artées, T Margalef, A Cortées - Procedia Computer Science, 2012 - Elsevier

… **prediction** quality improvements when coupling **forest fire spread model** with weather **forecast** … Rothermel fire **propagation model**[14], has been used as **forest fire spread** simulator and, …

☆ Save 99 Cite Cited by 34 Related articles All 6 versions

Kalman filter-based large-scale **wildfire** monitoring with a system of UAVs

Z Lin, HHT Liu, M Wotton - IEEE Transactions on Industrial …, 2018 - ieeexplore.ieee.org

… **wildfire propagation behavior** as well as fire front contour. It is enabled by developing a scalar field **wildfire model** … The dependence of **wildfire** models on the experiments and **survey** of …

☆ Save 99 Cite Cited by 150 Related articles All 2 versions

A **survey** of machine learning algorithms based forest fires **prediction** and detection systems

F Abid - Fire technology, 2021 - Springer

… the **behavior** of the **forest fire**, ie, forecasting the fire **spread** evolution. The **forest fire behavior prediction** … In this study a **model** for **wildfire** ignition probabilities **prediction** was proposed. A …

☆ Save 99 Cite Cited by 228 Related articles All 5 versions

Synthesis of knowledge of extreme fire **behavior**: volume I for fire managers

P Werth, B Potter, C Clements, M Finney, S Goodrick… - 2011 - scholarworks.sjsu.edu

… **review** of the scientific literature pertaining to extreme fire … by the National **Wildfire** Coordination Group makes the … starting point for enhanced **wildland fire management** and advances in …

☆ Save 99 Cite Cited by 154 Related articles All 18 versions »

Lessons learnt from post-**bushfire** surveys at the urban interface in Australia

R Blanchi, JE Leonard… - … Ecology and **Management**, 2006 - researchgate.net

… of ignition and **propagation** of **bushfire** in urban areas. The … **Survey** information covered by this paper is based on the CSIRO in-depth **survey** of the 1983 Ash Wednesday **bushfire** …

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Mapping Long-Term **Wildfire Dynamics** in Portugal Using **Trajectory** Analysis (1975–2024)

B Barbosa, A Gonçalves, S Oliveira, CM Viana - Land, 2025 - mdpi.com

… the first time, a **trajectory** analysis framework to **wildfire** occurrence data across … **wildfire** regimes and provide tools to support more adaptive and spatially differentiated fire **management** …

☆ Save 99 Cite Related articles All 9 versions »

Towards a comprehensive look at global drivers of novel extreme **wildfire** events

A Duane, M Castellnou, L Brotons - Climatic Change, 2021 - Springer

… **wildfire** events characterized by very high fire intensity and rapid rates of **spread**, and we **review** … to their process and **behavior**, together with a deficiency in current **simulation** models to …

☆ Save 99 Cite Cited by 327 Related articles All 10 versions

Efficient **simulation** of **wildfire spread** on an irregular grid

P Johnston, J Kelso, GJ Milne - … Journal of **Wildland Fire**, 2008 - CSIRO Publishing

… **Wildfire spread** is governed by several physical processes. The processes involved in the **spread** of a **wildfire** … Computational **simulation** of the physical processes is also difficult owing …

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Renormalization group approach to the critical **behavior** of the **forest-fire model**

V Loreto, L Pietronero, A Vespignani, S Zapperi - Physical **review** letters, 1995 - APS

… **forest-fire model** in order to characterize the nature of the critical state and its scale invariant **dynamics**.… This **model** is therefore critical in the usual sense because the control parameter …

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Wind effect on **wildfire** and **simulation** of its **spread** (case study: Siahkal forest in northern Iran).

R Jahdi, AA Darvishsefat, V Etemad, MA Mostafavi - 2014 - researchgate.net

… **Wildfire spread** and **behavior prediction** is important to bring the fire under control. This study validated the use of the FARSITE **model** in Northern Forests of Iran. The direction of **spread** …

☆ Save 99 Cite Cited by 30 Related articles All 62 versions »

Information systems in support of **wildland fire management** decision making in Canada

BS Lee, ME Alexander, BC Hawkes, TJ Lynham… - … and Electronics in …, 2002 - Elsevier

… on the CFFDRS as a means of predicting **wildland fire** occurrence and **behavior** potential. … These analyses can be used to support the setting or **review** of policy, allocation of national or …

☆ Save 99 Cite Cited by 273 Related articles All 8 versions

Effects of bark beetle-caused tree mortality on **wildfire**

JA Hicke, MC Johnson, JL Hayes… - … Ecology and **Management**, 2012 - Elsevier

… and crown fire **behavior** responses, leading to … **wildfire** such as climate. However, some disagreement among studies remained. Given the large areas of recent bark beetle and **wildfire** …

☆ Save 99 Cite Cited by 383 Related articles All 17 versions

Wind–terrain effects on the **propagation** of wildfires in rugged terrain: fire channelling

JJ Sharples, RHD McRae… - … Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… Implications of fire channelling for **bushfire** risk **management** are also discussed. … of the atypical fire **spread** as ‘fire-channelling’ events. By applying a simple terrain-filter **model** to the …

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Fuels and fire **behavior dynamics** in bark beetle-attacked forests in Western North America and implications for fire **management**

MJ Jenkins, WG Page, EG Hebertson… - … and **Management**, 2012 - Elsevier

… in forest health and **wildfire** protection policies of land **management** agencies. … **management** and threaten the safety of firefighters and the general public alike. In this paper, we **review** …

☆ Save 99 Cite Cited by 170 Related articles All 11 versions

Predicting the Future of Forest Fires: Climate Models, Fire **Behavior** and Fire **Spread**

P Mishra, H Singh, N Kumar - **Forest Fire** and Climate Change: Insights …, 2025 - Springer

… These models allow for the hourly **prediction** of **forest fire spread** after identifying the ignition … The **review** has flaws in everything from the coordination of preventative measures to the …

☆ Save 99 Cite Related articles All 3 versions

A **wildfire** risk assessment framework for land and resource **management**

JH Scott, MP Thompson, DE Calkin - 2013 - digitalcommons.unl.edu

… , geospatial information regarding the likely occurrence, **spread**, and consequences of **wildfire**. The use of burn probability … One such **survey** used in the past queried experts regard- …

☆ Save 99 Cite Cited by 310 Related articles All 9 versions »

How forest fires kill trees: a **review** of the fundamental biophysical processes

ST Michaletz, EA Johnson - Scandinavian Journal of Forest …, 2007 - Taylor & Francis

… Firsts, a more sophisticated fire **behavior model** is required so that the flaming fire front can … us to develop and improve our understanding of **forest fire behavior** and effects. K. Miyanishi, …

☆ Save 99 Cite Cited by 295 Related articles All 8 versions

Charts for interpreting **wildland fire behavior** characteristics

PL Andrews, RC Rothermel - 1982 - books.google.com

… As an illustration of how the fire **behavior** chart works, the fire **behavior prediction** values … A **review** of each index and the relationship between them will clarify the use of the NFDRS …

☆ Save 99 Cite Cited by 205 Related articles All 12 versions

AEGIS: a **wildfire** prevention and **management** information system

K Kalabokidis, A Ager, M Finney… - … Hazards and Earth …, 2016 - nhess.copernicus.org

… fire exposure to highly valued resources, and to **model** the effect of fuel breaks on **wildfire spread**. Under constant … and Ager, AA: A **review** of recent advances in risk analysis for **wildfire** …

☆ Save 99 Cite Cited by 87 Related articles All 14 versions »

Setting **wildfire** evacuation triggers by coupling fire and traffic **simulation** models: a spatiotemporal GIS approach

D Li, TJ Cova, PE Dennison - Fire technology, 2019 - Springer

… The results in this study demonstrate that we could better reveal the **dynamics** of evacuation traffic and fire **spread** in **wildfire** evacuations when we couple the two models to set triggers. …

☆ Save 99 Cite Cited by 97 Related articles All 7 versions

A **review** of the applications of remote sensing in fire ecology

DM Szpakowski, JLR Jensen - Remote sensing, 2019 - mdpi.com

… **wildfire spread**, wind direction, and weather conditions. An example of a commonly used **model** … FARSITE then uses existing fire **behavior** models to simulate how the fire will grow and …

☆ Save 99 Cite Cited by 320 Related articles All 9 versions »

Deep learning and transformer approaches for UAV-based **wildfire** detection and segmentation

R Ghali, MA Akhloufi, WS Mseddi - Sensors, 2022 - mdpi.com

… Early detection and **prediction** of fire **spread** can help reduce affected areas and improve firefighting. Numerous systems were developed to detect fire. Recently, Unmanned Aerial …

☆ Save 99 Cite Cited by 175 Related articles All 13 versions »

Self-organized critical **forest-fire model**

B Drossel, F Schwabl - Physical **review** letters, 1992 - APS

… , the **forest-fire** … the **model** rules, eg, anotherlattice symmetry or fire spreading to next-nearest neighbors. Let us compare the critical **behavior** of the **forest-fire model** to the sandpile **model**…

☆ Save 99 Cite Cited by 1006 Related articles All 10 versions

Incidence of **forest fire** in India and its effect on terrestrial ecosystem **dynamics**, nutrient and microbial status of soil

KK Chandra, AK Bhardwaj - International Journal of Agriculture and …, 2015 - academia.edu

… **management** of forest after fire will be more beneficial. The present **review** deals the impact of **forest fire** on species **dynamics**, … **management** and control of nutrient, insect and disease. …

☆ Save 99 Cite Cited by 104 Related articles All 3 versions »

A **review** of initial attack fire crew productivity and effectiveness

KG Hirsch, DL Martell - International Journal of **Wildland Fire**, 1996 - CSIRO Publishing

… and **spread** throughout the **forest fire management** community during the past 30 years. This information has been incorporated into a wide range of fire **management** … we **review** several …

☆ Save 99 Cite Cited by 171 Related articles All 8 versions

A real-time risk assessment tool supporting **wildland fire** decisionmaking

DE Calkin, MP Thompson, MA Finney… - Journal of …, 2011 - academic.oup.com

… fire **behavior** and **wildfire** impact modules of the **Wildland Fire** … fire **model** components, Fire **Spread** Probability (FSPro; fire … We **review** some real-world applications of WFDSS and show …

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Goat grazing as a **wildfire** prevention tool: A basic **review**

R Lovreglio, O Meddour-Sahar… - Iforest-Biogeosciences and …, 2014 - iforest.sisef.org

… to change **wildfire behavior**, by slowing its **spread**, shortening … The pastoralism, partner of **forest fire management**: 20 years … Europe, Forest **management** as a tool for **wildfire** prevention”. …

☆ Save 99 Cite Cited by 87 Related articles All 11 versions »

Rethinking the **wildland fire management** system

MP Thompson, DG MacGregor, CJ Dunn… - Journal of …, 2018 - academic.oup.com

… of **behavior**, and in so doing, challenge ingrained ways of thinking and acting that may be inconsistent with current intentions around **wildland fire management**. We … Such a **review** could …

☆ Save 99 Cite Cited by 144 Related articles All 13 versions

Time aware genetic algorithm for **forest fire propagation prediction**: exploiting multi-core platforms

T Artés, A Cencerrado, A Cortés… - Concurrency and …, 2017 - Wiley Online Library

… **forest fire** scenario and accomplishing it within strict time requirements to deliver a **forecast**, in this work, a complete **forest fire spread prediction** … accurate **propagation prediction** results …

☆ Save 99 Cite Cited by 38 Related articles All 4 versions

UAV patrolling for **wildfire** monitoring by a dynamic voronoi tessellation on satellite data

A Giuseppi, R Germanà, F Fiorini, F Delli Priscoli… - Drones, 2021 - mdpi.com

… The **prediction** of the **wildfire propagation** is a crucial functionality needed to enable any kind of proactive measure. For this reason, several models [22,23,24,25] have been proposed in …

☆ Save 99 Cite Cited by 38 Related articles All 5 versions »

Do thinning and/or burning treatments in western USA ponderosa or Jeffrey pine-dominated forests help restore natural fire **behavior**?

PZ Fulé, JE Crouse, JP Roccaforte, EL Kalies - … Ecology and **Management**, 2012 - Elsevier

… and burning treatments on restoring fire **behavior** attributes in western USA pine forests. … of **wildfire behavior** in treated stands have also been reported. Using a systematic **review** …

☆ Save 99 Cite Cited by 344 Related articles All 9 versions

A **review** of fire interactions and mass fires

MA Finney, SS McAllister - Journal of Combustion, 2011 - Wiley Online Library

… , including **wildland fire**, structural fire, as well as combustion engineering and fluid **dynamics**, … on interactions is directly relevant to **wildfire management** activities and tactics but has not …

☆ Save 99 Cite Cited by 142 Related articles All 7 versions

Fire **behavior** in masticated fuels: a **review**

JK Kreye, NW Brewer, P Morgan, JM Varner… - … and **Management**, 2014 - Elsevier

… Managing landscapes dependent on fire as an ecological process, while also mitigating the negative impacts of **wildfire** to people and property is challenging. In many fire-prone forest …

☆ Save 99 Cite Cited by 131 Related articles All 13 versions

Application of **wildfire** risk assessment results to **wildfire** response planning in the southern Sierra Nevada, California, USA

MP Thompson, P Bowden, A Brough, JH Scott… - Forests, 2016 - mdpi.com

… We **review** general … **wildfire** hazard across the landscape at a pixel size of 180 m (3 ha per pixel). FSim is a comprehensive fire occurrence, growth, **behavior**, and suppression **simulation** …

☆ Save 99 Cite Cited by 133 Related articles All 12 versions »

A GIS-supported **model** for the **simulation** of the spatial structure of **wildland fire**, Cass Basin, New Zealand

GLW Perry, AD Sparrow… - Journal of Applied Ecology, 1999 - Wiley Online Library

… **spread model** to New Zealand (NZ) ecosystems, and secondly, to assess the role GIS may play in fire **spread prediction**… to the data from the **survey** plots. Before dissection each whole (…

☆ Save 99 Cite Cited by 81 Related articles All 9 versions

Cooperative **forest fire** surveillance using a team of small unmanned air vehicles

DW Casbeer, DB Kingston, RW Beard… - International journal of …, 2006 - Taylor & Francis

… in **simulation** using a six degree-of-freedom dynamic **model** for the UAV and a numerical **propagation model** for the **forest fire**. … Let one UAV be assigned to **survey** the upper half of the …

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Self-autonomous wireless sensor nodes with wind energy harvesting for remote sensing of wind-driven **wildfire spread**

YK Tan, SK Panda - IEEE Transactions on Instrumentation and …, 2011 - ieeexplore.ieee.org

… coordinating the **review** process for … **Prediction** of the **Wildfire Spread** Based on the Wind Speed In the remote sensing of the **wildfire spread** condition, the **prediction** of the fire **behavior** …

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Coupling diagnostic and prognostic models to a dynamic data driven **forest fire spread prediction** system

C Brun, T Margalef, A Cortés - Procedia Computer Science, 2013 - Elsevier

… Finally, when considering the scheme where both prognostic and diagnostic models are coupled to the **forest fire spread model**, the **prediction spread** errors on average denote a great …

☆ Save 99 Cite Cited by 19 Related articles All 9 versions

Wildfire Impacts on Security of Electric Power Systems: A **Survey** of Risk Identification and Mitigation Approaches

R Shah, N Amjady, NSB Shakhawat, MU Aslam… - IEEE …, 2024 - ieeexplore.ieee.org

… The work in [44] has used a **wildfire prediction model** using … and recovery including **wildfire prediction** and network … to estimate the **wildfire dynamics** and **propagation** direction and its …

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Comparisons between **wildfire** and forest harvesting and their implications in forest **management**

DJ McRae, LC Duchesne, B Freedman… - Environmental …, 2001 - cdnsciencepub.com

… **review** we summarize empirical evidence that illustrates disparities between forest harvesting and **wildfire**… of disturbance by **wildfire** offer valuable insights into the ecological **dynamics** of …

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Wildland fire detection and monitoring using a drone-collected rgb/ir image dataset

X Chen, B Hopkins, H Wang, L O'Neill, F Afghah… - IEEE …, 2022 - ieeexplore.ieee.org

… leave uncertainties about a **wildfire**’s extent, **behavior**, and … of drones in remote **wildfire management** operations [12], [13]… Table 1 provides a **review** of existing fire datasets. As shown …

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Machine Learning based **Wildfire** Area Estimation Leveraging Weather **Forecast** Data

S Sultania, R Sonawane, P Kanikar - Int. J. Inf. Technol. Comput …, 2025 - mecs-press.org

… **wildfire prediction** to be used in a literature **review** as follows: … , improving **prediction** accuracy for **wildfire spread**. Bayesian … the inherent uncertainties of **wildfire prediction**. Reference [12…

☆ Save 99 Cite Cited by 1 Related articles All 2 versions »

Forest fire spread prediction using deep learning

F Khennou, J Ghaoui, MA Akhloufi - Geospatial informatics XI, 2021 - spiedigitallibrary.org

… , a deep learning **model** based on U-Net, past wildfires events and weather data. Our approach uses an intelligent **model** to study **forest fire spread** over a period of 24 hours. The **model** …

☆ Save 99 Cite Cited by 31 Related articles All 2 versions

Phoenix: development and application of a **bushfire** risk **management** tool

K Tolhurst, B Shields, D Chong - … of Emergency **Management** …, 2008 - search.informit.org

… measures for monitoring and **review** make up the risk **management** process as outlined in … the **Bushfire** Risk **Management Model** encapsulate the complex interaction of ignition, **spread**, …

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A **review** of challenges to determining and demonstrating efficiency of large fire **management**

MP Thompson, FR y Silva, DE Calkin… - … Journal of **Wildland Fire**, 2017 - CSIRO Publishing

Characterising the impacts of **wildland fire** and fire suppression is critical information for fire **management** decision-making. Here, we focus on decisions related to the rare larger and …

☆ Save 99 Cite Cited by 83 Related articles All 12 versions

Predicting crown fire **behavior** to support **forest fire management** decision-making

MG Cruz, ME Alexander, RH Wakimoto - … conference on **forest fire** …, 2002 - academia.edu

… **management**. This paper describes the modeling of two fundamental crown fire **behavior** features: the onset of crowning and the **spread** … of the Canadian **Forest Fire Behavior Prediction** …

☆ Save 99 Cite Cited by 88 Related articles All 5 versions »

The REBURN **model**: Simulating system-level forest succession and **wildfire dynamics**

SJ Prichard, RB Salter, PF Hessburg, NA Povak… - Fire Ecology, 2023 - Springer

… is the dominant driver of system-level vegetation **dynamics** within the Tripod landscape, the … REBURN **model** development and calibrated **model behavior**. We include a full **model** …

☆ Save 99 Cite Cited by 11 Related articles All 11 versions

Mapping **wildfire** susceptibility with the BURN-P3 **simulation model**

MA Parisien, VG Kafka, KG Hirsch, JB Todd, SG Lavoie… - 2005 - researchgate.net

… The **model** combines deterministic fire growth based on the Canadian Fire **Behavior Prediction** … probabilistic fire ignitions and **spread** events derived from historical fire and weather data. …

☆ Save 99 Cite Cited by 153 Related articles All 2 versions »

Advanced automatic **wildfire** surveillance and monitoring network

D Stipaničev, M Štula, D Krstinić, L Šerić… - … on **Forest Fire** …, 2010 - academia.edu

… In this paper, after a short **survey** of various approaches to the **wildfire** monitoring, … **management** activities. GIS data is important for fire **behavior** modeling, **forest fire spread simulation** …

☆ Save 99 Cite Cited by 52 Related articles All 5 versions »

Wildfire policy and public lands: integrating scientific understanding with social concerns across landscapes

MP Dombeck, JE Williams, CA Wood - Conservation biology, 2004 - Wiley Online Library

… **wildfire** in the United States and our attempts to control it contain many lessons to guide future policies for **management** … weeds **spread** at a rate of 688,500 ha annually (Bureau of Land …

☆ Save 99 Cite Cited by 288 Related articles All 13 versions

A real-time web-based **wildfire simulation** system

R Wu, C Chen, S Ahmad, JM Volk… - IECON 2016-42nd …, 2016 - ieeexplore.ieee.org

… Therefore, a simplified **model** of fire **propagation simulation** was developed in this study. … taken intermittingly by the United States Geological **Survey** (USGS) between 1948 and 2010. …

☆ Save 99 Cite Cited by 25 Related articles All 9 versions

Interpreting and using outputs from the Canadian **Forest Fire** Danger Rating System in research applications

BM Wotton - Environmental and ecological statistics, 2009 - Springer

… **review** will expose statistical modellers and other researchers to some of the models used currently in **forest fire management** … quantities such as rate of **spread**, fuel consumption and …

☆ Save 99 Cite Cited by 334 Related articles All 12 versions

Automatic **wildfire** detection and **simulation** using optical information from unmanned aerial systems

CW Lum, A Summers, B Carpenter, A Rodriguez… - 2015 - sae.org

… are simply not able to **survey** the amount of area necessary … (UAS) for automated **wildfire** detection. The proposed system … for the detection and **simulation** of fire **propagation** from UAS …

☆ Save 99 Cite Cited by 13 Related articles All 4 versions

Spatial-temporal distribution **prediction** of transmission corridor **wildfire** risk based on ARIMA-DBN

E Zhou, L Wang, H Wang, Z Wang… - Frontiers in Forests and …, 2025 - frontiersin.org

… **model** for assessing the spatiotemporal risk of **wildfire** occurrence in transmission corridors, with an emphasis on the role of meteorological factors in short-term **wildfire dynamics**. A …

☆ Save 99 Cite Related articles All 3 versions »

Integrated **wildfire** risk assessment: Framework development and application on the Lewis and Clark National Forest in Montana, USA

MP Thompson, J Scott, D Helmbrecht… - … and **management**, 2013 - academic.oup.com

… consider biophysical variables beyond fire **behavior**. We anticipate that this work will further the state of … Below we briefly **review** each step in the context of the Little Belts Assessment. …

☆ Save 99 Cite Cited by 84 Related articles All 17 versions

Wildfire management in the United States: the evolution of a policy failure

G Busenberg - **Review** of policy research, 2004 - Wiley Online Library

… **model** of the origins and **propagation** of policy failures. The elements of this policy failure **model** … This article applies the policy failure **model** to explore the evolution of US **wildfire** policy, …

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Prediction of **forest fire spread** rate using UAV images and an LSTM **model** considering the interaction between fire and wind

X Li, H Gao, M Zhang, S Zhang, Z Gao, J Liu, S Sun… - Remote Sensing, 2021 - mdpi.com

… **model** to predict the **spread** rate of the **forest fire**. In order to deeply capture the characteristics of **forest fire spread** by … impact to the process of **forest fire spread** as the input parameters to …

☆ Save 99 Cite Cited by 35 Related articles All 8 versions »

A Comparative Study of a Deep Reinforcement Learning Solution and Alternative Deep Learning Models for **Wildfire Prediction**.

C Vidal-Silva, R Pizarro… - Applied Sciences …, 2025 - search.ebscohost.com

… dependencies in **wildfire dynamics**. The findings can support the integration of deep learning models into early warning systems, contributing to proactive **wildfire** risk **management**. …

☆ Save 99 Cite Cited by 3 Related articles All 2 versions

Trying not to get burned: Understanding homeowners' **wildfire** risk–mitigation behaviors

H Brenkert-Smith, PA Champ, N Flores - Environmental **Management**, 2012 - Springer

… WUI property owners about **wildfire** risks. In this study, we analyze **survey** data from WUI … to the current literature on homeowners and **wildfire** risk–mitigation **behavior**. First, we move …

☆ Save 99 Cite Cited by 214 Related articles All 18 versions

LinU-Mamba: Visual Mamba U-Net with Linear Attention to Predict **Wildfire Spread**

HS Andrianarivony, MA Akhloufi - Remote Sensing, 2025 - mdpi.com

… In this study, we develop a deep learning **model** to predict **wildfire spread** using remote … Collectively, these deep models deliver the highest performance reported in the **review**, but the …

☆ Save 99 Cite Related articles All 5 versions »

Cloud-based urgent computing for **forest fire spread prediction**

E Fraga, A Cortés, T Margalef, P Hernández… - … Modelling & Software, 2024 - Elsevier

… **Forest fire simulation** have improved significantly, but input data describing fire scenarios are subject to high levels of uncertainty. In this work the two-stage **prediction** … actual **prediction** …

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A comprehensive analysis of the effect of radiation and Sherwood numbers in **wildfire model** with moisture convection and diffusion mechanisms

AB Zhiri, OA Ajala, AO Akindele, P Adegbite… - Journal of Radiation …, 2025 - Elsevier

… The **spread** of forest fires in this study has been modeled using a … **model** offers a valuable tool for researchers to investigate how climate change may influence future **wildfire behavior**…

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A **review** of **forest fire** surveillance technologies: Mobile ad-hoc network routing protocols perspective

FT Al-Dhief, N Sabri, S Fouad, NMA Latiff… - Journal of King Saud …, 2019 - Elsevier

… and developers for various applications such as **forest fire** detection. Forest fires require continuous … This paper gives an extensive **review** of the modern techniques used in the **forest fire** …

☆ Save 99 Cite Cited by 84 Related articles All 2 versions

Wildfiredb: An open-source dataset connecting **wildfire spread** with relevant determinants

S Singla, A Mukhopadhyay, M Wilbur, T Diao… - Conference on Neural …, 2021 - cs.ucr.edu

… Our goal is to present a data source that can be used to **model** both fire occurrences and the **spread** of fires. We discretize space and time to create our dataset (we discuss the unit of …

☆ Save 99 Cite Cited by 12 Related articles »

Mapping regional **forest fire** probability using artificial neural network **model** in a Mediterranean forest ecosystem

O Satir, S Berberoglu, C Donmez - Geomatics, Natural Hazards …, 2016 - Taylor & Francis

… In this paper, the basic feedforward, back-**propagation** ANN described above is used as a regression **model** to estimate fire probability based on input features that include the climate, …

☆ Save 99 Cite Cited by 247 Related articles All 8 versions

Coupled atmosphere-**wildland fire** modeling with WRF-Fire version 3.3

J Mandel, JD Beezley… - Geoscientific **Model** …, 2011 - gmd.copernicus.org

… **wildland fire behavior** can be captured by the coupling of a mesoscale weather **model** with a simple 2-D fire **spread model** (… Sullivan, AL: A **review** of **wildland fire spread** modelling, 1990…

☆ Save 99 Cite Cited by 361 Related articles All 18 versions »

Fire intensity, fire severity and burn severity: a brief **review** and suggested usage

JE Keeley - International journal of **wildland fire**, 2009 - CSIRO Publishing

… This metric is most useful in understanding fire **behavior** in … precise metric varies with **management** needs. Confusion arises … of these terms and suggest a **model** that may help clarify …

☆ Save 99 Cite Cited by 2613 Related articles All 15 versions

A decision-making framework for **wildfire** suppression

N Petrovic, JM Carlson - International Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… fire **spread model**, an economic cost **model** and population data. Our study uses **model**-… multiple fires using scenarios motivated by the 2003 and 2007 California **wildfire** seasons. …

☆ Save 99 Cite Cited by 23 Related articles All 5 versions

The Rothermel surface fire **spread model** and associated developments: A comprehensive explanation

PL Andrews - Gen. Tech. Rep. RMRS-GTR-371. Fort Collins …, 2018 - research.fs.usda.gov

… I hope that this paper will serve as a valuable reference for students of **wildland fire** … Sullivan (2009a,b,c) did a comprehensive **review** of wildland surface fire **spread** models developed …

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Simulation of a large **wildfire** in a coupled fire-atmosphere **model**

JB Filippi, F Bosseur, C Mari, C Lac - Atmosphere, 2018 - mdpi.com

… fire **spread model** is proposed for predicting fine- to large-scale effects of this extreme **wildfire**, showing that such **simulation** is … (see [31] for a **review** of numerical interface tracking and …

☆ Save 99 Cite Cited by 53 Related articles All 10 versions »

Integrated national-scale assessment of **wildfire** risk to human and ecological values

MP Thompson, DE Calkin, MA Finney, AA Ager… - … Research and Risk …, 2011 - Springer

… **wildfire** entailed use of FSim, a **wildfire simulation** system. The specifics of FSim, including details on how fire **behavior** … impacted by fire) through **review** of many recovery plans and …

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Wildfire risk in the wildland–urban interface: a **simulation** study in northwestern Wisconsin

AB Massada, VC Radeloff, SI Stewart… - … and **Management**, 2009 - Elsevier

… We simulated **wildfire spread** and **behavior** with two commonly used models: FARSITE (Finney, 1998) and MTT (Finney, 2002), as implemented in FlamMap (Finney, 2006). FARSITE is …

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Homeowners willingness to pay to reduce **wildfire** risk in wildland urban interface areas: Implications for targeting financial incentives

JJ Sanchez, TP Holmes, J Loomis… - International Journal of …, 2022 - Elsevier

… This suggests the importance of risk perceptions in adoption of risk mitigation **behavior**. … **wildfire** risk perceptions and mitigation **behavior**, not explicitly revealed in the basic MNL **model**, …

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Socio-Ecological Perceptions of **Wildfire Management** and Effects in Greece. Fire 2021, 4, 18

P Palaiologou, K Kalabokidis, A Troumbis, MA Day… - 2021 - academia.edu

… emphasis on reforming **wildfire management** policies to deal … of climate change in **wildfire** frequency and **behavior**. In addition… -based **survey** of professionals in **wildfire management** and …

☆ Save 99 Cite Cited by 8 Related articles All 2 versions »

An integrative **review** of empirical research on perceptions and behaviors related to prescribed burning and **wildfire** in the United States

LN Dupéy, JW Smith - Environmental **management**, 2018 - Springer

… (evacuation), also increased the likelihood of engaging in mitigation **behavior**. These two studies highlight how emerging research is beginning to explore a wider collection of …

☆ Save 99 Cite Cited by 52 Related articles All 9 versions

Model comparisons for estimating carbon emissions from North American **wildland fire**

NHF French, WJ De Groot, LK Jenkins… - Journal of …, 2011 - Wiley Online Library

… by the sixth **model** used in this study, the Canadian **Forest Fire Behavior Prediction** (FBP) … Here we focus our **review** on total carbon emissions estimates from burning, with the aim of …

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The role of UAV-IoT networks in future **wildfire** detection

OM Bushnaq, A Chaaban… - IEEE Internet of Things …, 2021 - ieeexplore.ieee.org

… In [17], a **survey** on the flying ad hoc networks is presented. The … , we assume a simplistic fire **spread model**, wherein fire evolves … In Section II, we present the system **model** and the main …

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Exploring adoption of the **wildland fire** decision support system: End user perspectives

P Noble, TB Paveglio - Journal of Forestry, 2020 - academic.oup.com

… WFDSS is a web-based application that synthesizes geospatial data, fire **behavior model** outputs, risk analysis frameworks, and cost evaluation tools. It is intended to help fire managers …

☆ Save 99 Cite Cited by 56 Related articles All 9 versions

Social amplification of **wildfire** risk: the role of social interactions and information sources

H Brenkert-Smith, KL Dickinson, PA Champ… - Risk …, 2013 - Wiley Online Library

… , we develop a conceptual **model** of **wildfire** risk perceptions that … To the extent that people are aware of these **dynamics**, we … Here, we briefly describe the **survey** design, as well as the …

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Psychological factors and social processes influencing **wildfire** smoke protective **behavior**: Insights from a case study in Northern California

FN Santana, DJX Gonzalez, G Wong-Parodi - Climate Risk **Management**, 2021 - Elsevier

… health **behavior**. Finally, we propose a conceptual framework of **wildfire** smoke protective **behavior**. … risk and protective health literatures related to **wildfire** smoke response. Our findings …

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Data assimilation for wildland fires

J Mandel, JD Beezley, JL Coen… - IEEE Control Systems …, 2009 - ieeexplore.ieee.org

… A **wildland fire** interacts with the atmosphere **dynamics** through fluxes of momentum, water … Then the state of the **model** is the probability distribution of possible **wildfire** scenarios. The …

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Simulation of **forest fire** fronts using cellular automata

AH Encinas, LH Encinas, SH White, AM Del Rey… - … in Engineering Software, 2007 - Elsevier

… Specifically, the **model** proposed in this work introduces a more accurate factor of **propagation** from a diagonal neighbor cell and includes, in a detailed form, the rate of fire **spread**. …

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Spatiotemporal **dynamics** of simulated **wildfire**, forest **management**, and forest succession in central Oregon, USA

AMG Barros, AA Ager, MA Day, HK Preisler, TA Spies… - Ecology and …, 2017 - JSTOR

… **simulation model** Envision to analyze long-term **wildfire dynamics** and the effects of different fuel **management** … We simulated a 50-year future where fuel **management** activities were …

☆ Save 99 Cite Cited by 49 Related articles All 11 versions

Forest fire spread model based on the grey system theory.

C Lv, J Wang, F Zhang - Journal of Supercomputing, 2020 - search.ebscohost.com

… **forest fire spread model** developing. By preprocessing and fusing the MODIS remote sensing data, the sequence data of the grey **model** … algorithm, **forest fire spread** GM(1, 1) **model** was …

☆ Save 99 Cite Cited by 12 Related articles All 2 versions

Comparing production function models for **wildfire** risk analysis in the wildland–urban interface

DE Mercer, JP Prestemon - Forest policy and economics, 2005 - Elsevier

… to **wildfire** … **wildland fire management** decisions in the wildland–urban interface. Our results also emphasize the importance of including such variables in statistical models of **wildfire** risk …

☆ Save 99 Cite Cited by 166 Related articles All 14 versions

Carbon **dynamics** of North American boreal forest after stand replacing **wildfire** and clearcut logging

M Seedre, BM Shrestha, HYH Chen… - Journal of forest …, 2011 - Taylor & Francis

… this **review** was to synthesize the literature on C **dynamics** of … disturbances, stand replacing **wildfire** and clearcut logging. … of **wildfire** and harvesting stands with different **management** …

☆ Save 99 Cite Cited by 98 Related articles All 20 versions

Conflict and collaboration in **wildfire management**: the role of mission alignment

CJ Fleming, EB McCartha… - … Administration **Review**, 2015 - Wiley Online Library

… By understanding where differences can arise and whether they are managed well, we provide insight into **dynamics** of relationships that will need to be examined and addressed. …

☆ Save 99 Cite Cited by 95 Related articles All 9 versions

Health effects of **wildfire** smoke exposure

CF Gould, S Heft-Neal, M Johnson… - Annual **Review** of …, 2024 - annualreviews.org

… and social systems adapt to new **dynamics** (eg, increased **wildfire** activity) by changing aspects of … smoke days have low to medium PM2.5 concentrations, using the quartic **model** we …

☆ Save 99 Cite Cited by 118 Related articles All 6 versions

Toward improved communication about **wildland fire**: mental models research to identify information needs for natural resource **management**

M Zaksek, JL Arvai - Risk Analysis: An International Journal, 2004 - Wiley Online Library

… **model** was created based on an extensive **review** of the literature and consultations with several **wildland fire** … as factors that influence **wildland fire** ignition and **spread**. Similarly, only …

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